Project

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Background

Data Cleaning

```
## Warning: package 'sfsmisc' was built under R version 4.5.2
```

```
##   X      player_name team_abbreviation age player_height player_weight
## 1 0 Randy Livingston               HOU  22        193.04      94.80073
## 2 1 Gaylon Nickerson               WAS  28        190.50      86.18248
## 3 2     George Lynch               VAN  26        203.20     103.41898
## 4 3   George McCloud               LAL  30        203.20     102.05820
## 5 4     George Zidek               DEN  23        213.36     119.74829
## 6 5   Gerald Wilkins               ORL  33        198.12     102.05820
##                 college country draft_year draft_round draft_number gp  pts reb
## 1       Louisiana State     USA       1996           2           42 64  3.9 1.5
## 2 Northwestern Oklahoma     USA       1994           2           34  4  3.8 1.3
## 3        North Carolina     USA       1993           1           12 41  8.3 6.4
## 4         Florida State     USA       1989           1            7 64 10.2 2.8
## 5                  UCLA     USA       1995           1           22 52  2.8 1.7
## 6 Tennessee-Chattanooga     USA       1985           2           47 80 10.6 2.2
##   ast net_rating oreb_pct dreb_pct usg_pct ts_pct ast_pct  season
## 1 2.4        0.3    0.042    0.071   0.169  0.487   0.248 1996-97
## 2 0.3        8.9    0.030    0.111   0.174  0.497   0.043 1996-97
## 3 1.9       -8.2    0.106    0.185   0.175  0.512   0.125 1996-97
## 4 1.7       -2.7    0.027    0.111   0.206  0.527   0.125 1996-97
## 5 0.3      -14.1    0.102    0.169   0.195  0.500   0.064 1996-97
## 6 2.2       -5.8    0.031    0.064   0.203  0.503   0.143 1996-97
```

```
## [1] 12844    22
```

```
##       X   player_name team_abbreviation age player_height player_weight
## 1 12590    AJ Griffin               ATL  19        198.12      99.79024
## 2  9405    AJ Hammons               DAL  24        213.36     117.93392
## 3 10270 Aaron Holiday               IND  22        185.42      83.91452
## 4 11332 Aaron Nesmith               BOS  21        195.58      97.52228
## 5 11959 Aaron Wiggins               OKC  23        193.04      86.18248
## 6  9922   Abdel Nader               BOS  24        198.12     104.32616
##      college country draft_year draft_round draft_number gp pts reb ast
## 1       Duke     USA       2022           1           16 72 8.9 2.1 1.0
## 2     Purdue     USA       2016           2           46 22 2.2 1.6 0.2
## 3       UCLA     USA       2018           1           23 50 5.9 1.3 1.7
```

```
## 4 Vanderbilt     USA       2020           1           14 46 4.7 2.8 0.5
## 5   Maryland     USA       2021           2           55 50 8.3 3.6 1.4
## 6 Iowa State   Egypt       2016           2           58 48 3.0 1.5 0.5
##   net_rating oreb_pct dreb_pct usg_pct ts_pct ast_pct  season
## 1        1.5    0.026    0.080   0.174  0.577   0.070 2022-23
## 2       -0.6    0.049    0.199   0.167  0.472   0.038 2016-17
## 3        7.0    0.008    0.088   0.206  0.518   0.180 2018-19
## 4       -0.5    0.041    0.146   0.133  0.573   0.047 2020-21
## 5       -8.1    0.040    0.098   0.149  0.556   0.083 2021-22
## 6      -10.3    0.024    0.111   0.162  0.439   0.078 2017-18
```

```
suppressPackageStartupMessages(require(ggplot2))
suppressPackageStartupMessages(require(dplyr))
suppressPackageStartupMessages(require(tidyverse))
suppressPackageStartupMessages(require(broom))

rookie_data$draft_round <- ifelse(rookie_data$draft_round==1,1,0)
```

EDA

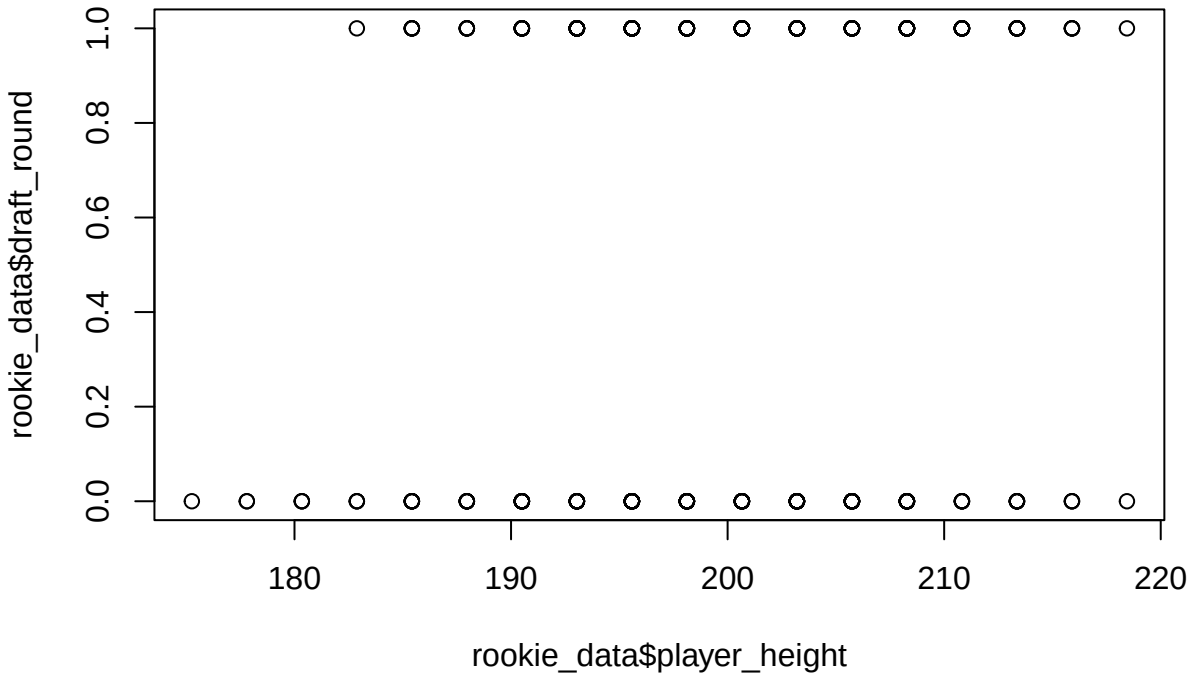
This is the section for more graphs, observed patterns and interpretations.

```
glm(formula = draft_round~player_height,
       data = rookie_data,
       family = binomial)%>%
    tidy(exponentiate = F)
```

```
## # A tibble: 2 x 5
##   term          estimate std.error statistic p.value
##   <chr>            <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)    -5.35      2.54       -2.11  0.0351
## 2 player_height   0.0276    0.0127      2.17  0.0298
```

In the frequentist interpretation, since the p value for slope is smaller than 0.05 (0.029), and the slope is larger than 0 (0.027), we could conclude that there is a statistically significant positive association between the player's height and draft round. We could like to also verify this idea using the Bayesian approach.

```
plot(y=rookie_data$draft_round,x=rookie_data$player_height)
```



From the scatter plot, we can see that when player's height is below 185cm, it is very unlikely for the player to be drafted in the first round. Howeverm after 185cm, the nuisance becomes less obvious.

The model has distribution as following.

$$\textbf{slope} \sim N(0, 10)$$

$$\textbf{intercept} \sim N(0, 10)$$

$$\theta(\textbf{height}) = logistic(\textbf{intercept+height*slope})$$

$$Y \sim Bernoulli(\theta(\textbf{height}))$$

Model Fitting

```
suppressPackageStartupMessages(library(cmdstanr))
set.seed(405)

data_list <- list(
  N = nrow(rookie_data),
  height = rookie_data$player_height,
```

```r
  y = rookie_data$draft_round
)

mod <- cmdstan_model("logistic_slope.stan")

fit <- mod$sample(
  data = data_list,
  seed = 1,
  chains = 4,
  iter_sampling = 5000,
  refresh = 0
)
```

```
## Running MCMC with 4 sequential chains...

## Chain 1 finished in 4.5 seconds.
## Chain 2 finished in 4.8 seconds.

## Chain 3 finished in 4.5 seconds.

## Chain 4 finished in 4.4 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 4.6 seconds.
## Total execution time: 18.5 seconds.
```

```r
fit
```

```
##        variable    mean  median   sd  mad      q5     q95 rhat ess_bulk
##  lp__           -269.13 -268.83 0.98 0.70 -271.11 -268.20 1.00     4652
##  intercept        -5.08   -5.07 2.45 2.45   -9.19   -1.06 1.00     2512
##  slope             0.03    0.03 0.01 0.01    0.01    0.05 1.00     2514
##  slope_positive    0.98    1.00 0.12 0.00    1.00    1.00 1.00     2650
##  ess_tail
##      5153
##      2645
##      2679
##        NA
```

```r
cat("Exponentiated value of the slope coefficient is", exp(mean(fit$draws("slope"))),".")
```

```
## Exponentiated value of the slope coefficient is 1.026598 .
```

We choose to use 4 chains so we could check whether the chain mixes well later. From the table, we can conclude that the mean of slope is 0.03, and the mean of `slope_positive`, an indicator variable is 0.98.

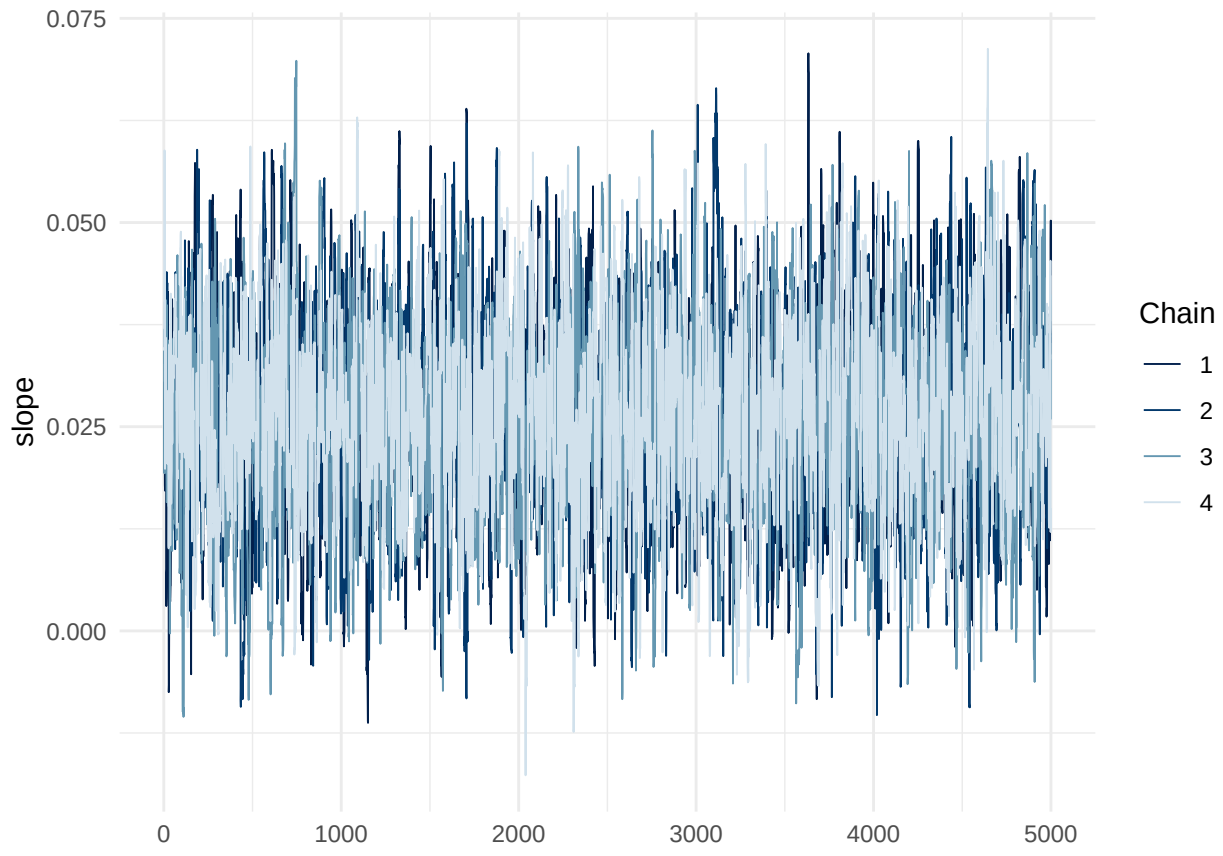
Since $E[1[slope > 0]|Y] = P(slope > 0|Y)$, this model effectively answered our question, as the probability of interest is substancially larger than 0.5. Thus, we conclude that height has a positive relationship with the chance of being drafted in the first round. Thus, being taller in height means higher probability of drafted earlier.

From the table, we can also conclude that a 1 cm increase in height is associated with a 2.6% increase in the odds of a player being drafted in the first round.

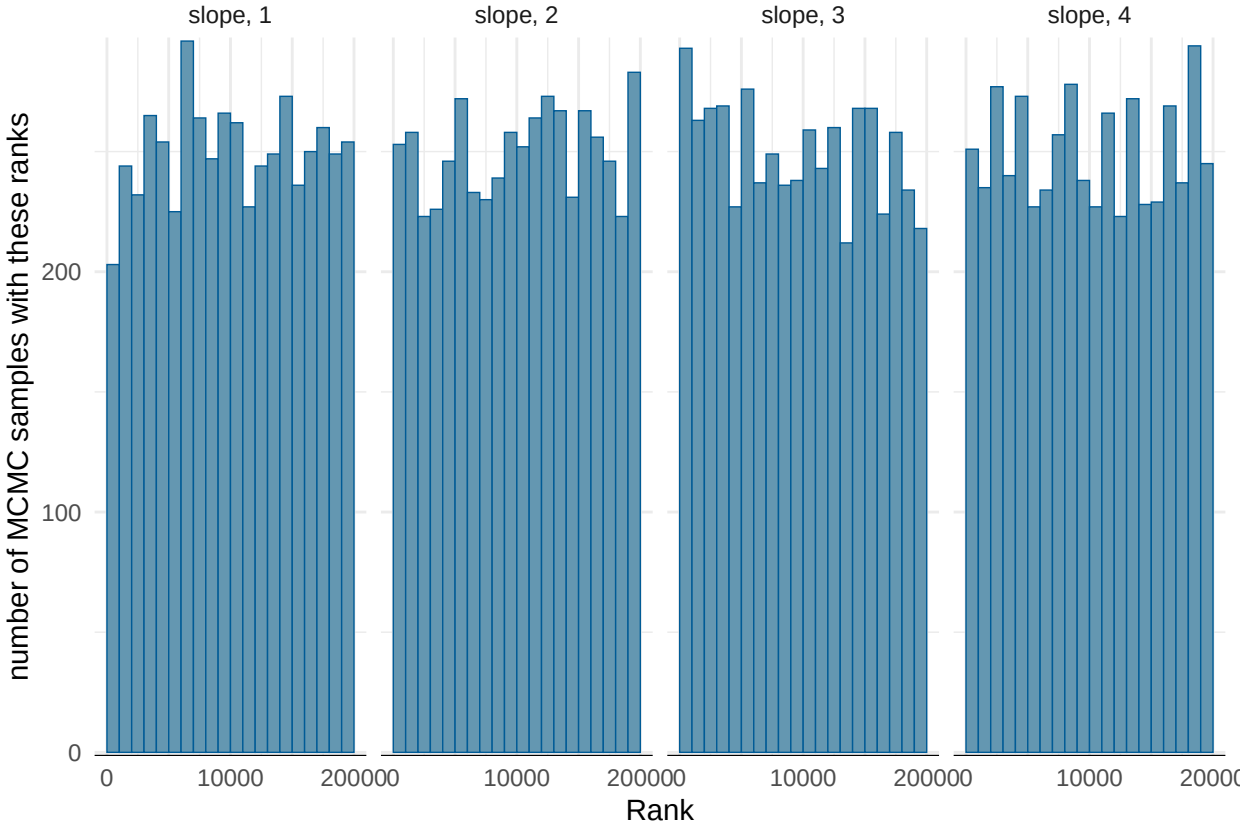
We would like to assess the goodness of fit using the following approach.

MCMC Diagnostics

```
suppressPackageStartupMessages(library(bayesplot))
mcmc_trace(fit$draws("slope")) + theme_minimal()
```




```
mcmc_rank_hist(fit$draws("slope")) +
  ylab("number of MCMC samples with these ranks") +
  theme_minimal()+
  scale_x_continuous(
  breaks = c(0, 5000, 10000, 15000, 20000),
  labels = c("0", "", "10000", "", "20000")
)
```



Here, we are using `slope` instead of `slope_positive` to check whether the chain mixes well. this is because that `slope` is a continuous variable , but `slope_positive` is binary. Thus, fitting the trace plots and histograms on this binary variable will most likely generate some vertical lines, which becomes harder to interpret. Thus, we use `slope` instead. Since both variables are form the same model, if the plots for `slope` suggests fast mixing, then it means that the model for `slope_positive` mixes fast.

From the trace plot, we can see that they four chains overlap a lot; from the histogram, we can see four approximately uniform histograms. Both plots suggests that our model is fast mixing.

Goodness of Fit

this is the section for calibration